

Algebra II with Trigonometry

Chapter 6.3

Olympiad Solutions (GPT-5.4)

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Solutions

1. Find the exact maximum value of

$$\sin x + \cos x + \sin 2x$$

for real x .

Let

$$S = \sin x + \cos x + \sin 2x.$$

Using $\sin 2x = 2 \sin x \cos x$, write

$$S = \sin x + \cos x + 2 \sin x \cos x.$$

Set

$$u = \sin x + \cos x.$$

Then

$$u^2 = \sin^2 x + \cos^2 x + 2 \sin x \cos x = 1 + \sin 2x,$$

so

$$\sin 2x = u^2 - 1.$$

Hence

$$S = u + (u^2 - 1) = u^2 + u - 1.$$

Now

$$u = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}].$$

Since $u^2 + u - 1$ is an upward-opening quadratic, its maximum on the interval $[-\sqrt{2}, \sqrt{2}]$ occurs at an endpoint. Evaluate:

$$S(\sqrt{2}) = 2 + \sqrt{2} - 1 = 1 + \sqrt{2},$$

$$S(-\sqrt{2}) = 2 - \sqrt{2} - 1 = 1 - \sqrt{2}.$$

Therefore the maximum is

$$\boxed{1 + \sqrt{2}}.$$

This occurs when $\sin x + \cos x = \sqrt{2}$, i.e. at

$$x = \frac{\pi}{4} + 2\pi k.$$

2. A sinusoid of the form

$$y = a \cos(kx - b) + c$$

has one complete period on $[1, 5]$, maximum value 7, minimum value -1 , and passes through $(1, 3)$ and $(3, 7)$. Determine one such equation.

From the maximum and minimum values,

$$a = \frac{7 - (-1)}{2} = 4, \quad c = \frac{7 + (-1)}{2} = 3.$$

So the function has the form

$$y = 4 \cos(kx - b) + 3.$$

Since one complete period occurs on $[1, 5]$, the period is

$$T = 5 - 1 = 4.$$

For $y = \cos(kx - b)$, the period is $2\pi/k$, so

$$\frac{2\pi}{k} = 4 \quad \Rightarrow \quad k = \frac{\pi}{2}.$$

Thus

$$y = 4 \cos \left(\frac{\pi x}{2} - b \right) + 3.$$

Now use the point $(3, 7)$. Since 7 is the maximum,

$$4 \cos \left(\frac{3\pi}{2} - b \right) + 3 = 7$$

gives

$$\cos \left(\frac{3\pi}{2} - b \right) = 1.$$

Hence

$$\frac{3\pi}{2} - b = 2\pi n.$$

Take $n = 0$, so

$$b = \frac{3\pi}{2}.$$

Then

$$y = 4 \cos \left(\frac{\pi x}{2} - \frac{3\pi}{2} \right) + 3.$$

Check $(1, 3)$:

$$y(1) = 4 \cos \left(\frac{\pi}{2} - \frac{3\pi}{2} \right) + 3 = 4 \cos(-\pi) + 3 = -4 + 3 = -1,$$

so this does not pass through $(1, 3)$. Thus we should interpret $(3, 7)$ as a maximum point, but the phase must also fit $(1, 3)$, so let us instead use both

points directly.

From (1, 3):

$$4 \cos \left(\frac{\pi}{2} - b \right) + 3 = 3 \quad \Rightarrow \quad \cos \left(\frac{\pi}{2} - b \right) = 0.$$

So

$$\frac{\pi}{2} - b = \frac{\pi}{2} + n\pi \quad \text{or} \quad -b = n\pi,$$

hence

$$b = n\pi.$$

From (3, 7):

$$4 \cos \left(\frac{3\pi}{2} - b \right) + 3 = 7 \quad \Rightarrow \quad \cos \left(\frac{3\pi}{2} - b \right) = 1.$$

Thus

$$\frac{3\pi}{2} - b = 2m\pi.$$

If $b = n\pi$, then

$$\frac{3\pi}{2} - n\pi = 2m\pi,$$

which has no integer solution. So with $a = 4$, no solution arises.

Hence we may take $a = -4$ instead, since the amplitude is $|a| = 4$. Then

$$y = -4 \cos \left(\frac{\pi x}{2} - b \right) + 3.$$

Use (3, 7):

$$-4 \cos \left(\frac{3\pi}{2} - b \right) + 3 = 7 \quad \Rightarrow \quad \cos \left(\frac{3\pi}{2} - b \right) = -1.$$

So

$$\frac{3\pi}{2} - b = (2m + 1)\pi.$$

Take $m = 0$, giving

$$b = \frac{\pi}{2}.$$

Then

$$y = -4 \cos \left(\frac{\pi x}{2} - \frac{\pi}{2} \right) + 3.$$

Check $(1, 3)$:

$$y(1) = -4 \cos 0 + 3 = -1,$$

still not correct.

So let us shift the maximum to $x = 3$ and require one full period from $x = 1$ to $x = 5$. Since $x = 1$ and $x = 5$ are corresponding points one period apart, and $y(1) = 3$, $y(5) = 3$ as well. Also $x = 3$ is halfway through the period from 1 to 5, so the phase should move by π from $x = 1$ to $x = 3$. Since $y(1) = 3$ is the midline and $y(3) = 7$ is the maximum, a sine form is natural:

$$y = 4 \sin \left(\frac{\pi}{2}(x - 1) \right) + 3.$$

Convert to cosine:

$$\sin \theta = \cos \left(\theta - \frac{\pi}{2} \right),$$

so

$$y = 4 \cos \left(\frac{\pi}{2}(x - 1) - \frac{\pi}{2} \right) + 3 = 4 \cos \left(\frac{\pi x}{2} - \pi \right) + 3.$$

Check:

- At $x = 1$: $\cos(-\pi) = -1$, so $y = -4 + 3 = -1$, not 3.

So instead we want the graph to start on the midline and rise to a maximum after 2 units. Thus

$$y = 4 \sin \left(\frac{\pi}{2}(x - 1) \right) + 3$$

indeed gives

$$y(1) = 3, \quad y(3) = 3,$$

so that also fails.

To reach a maximum at $x = 3$, we need

$$\frac{\pi}{2}(3 - 1) = \pi,$$

which is not where sine is maximal.

A better choice is to place cosine maximum at $x = 3$:

$$y = 4 \cos \left(\frac{\pi}{2}(x - 3) \right) + 3.$$

Then

$$y(3) = 7,$$

and

$$y(1) = 4 \cos(-\pi) + 3 = -1.$$

Again not correct.

Thus $x = 3$ cannot be a maximum if the period is exactly 4 and $x = 1$ is a midline crossing; the data are inconsistent unless “one complete period on $[1, 5]$ ” means only that the interval contains a period, not necessarily starting at matching phase endpoints.

Take period 4, so $k = \pi/2$, and solve:

$$4 \cos \left(\frac{\pi}{2} - b \right) + 3 = 3 \Rightarrow \cos \left(\frac{\pi}{2} - b \right) = 0,$$

so

$$b = n\pi.$$

Also

$$4 \cos \left(\frac{3\pi}{2} - b \right) + 3 = 7 \Rightarrow \cos \left(\frac{3\pi}{2} - b \right) = 1.$$

This has no solution. With $a = -4$,

$$-4 \cos \left(\frac{\pi}{2} - b \right) + 3 = 3 \Rightarrow \cos \left(\frac{\pi}{2} - b \right) = 0,$$

again $b = n\pi$, and

$$-4 \cos \left(\frac{3\pi}{2} - b \right) + 3 = 7 \Rightarrow \cos \left(\frac{3\pi}{2} - b \right) = -1,$$

which also has no solution.

Therefore the given data are inconsistent; **no such sinusoid exists.**

There is no equation of the stated form satisfying all conditions.

3. Compute exactly:

$$\sin \frac{\pi}{12} + \sin \frac{5\pi}{12} + \sin \frac{9\pi}{12} + \sin \frac{13\pi}{12} + \sin \frac{17\pi}{12}.$$

Write the angles in degrees:

$$15^\circ, 75^\circ, 135^\circ, 195^\circ, 255^\circ.$$

So the sum is

$$\sin 15^\circ + \sin 75^\circ + \sin 135^\circ + \sin 195^\circ + \sin 255^\circ.$$

Now use

$$\sin(180^\circ + \theta) = -\sin \theta.$$

Thus

$$\sin 195^\circ = -\sin 15^\circ, \quad \sin 255^\circ = -\sin 75^\circ.$$

These cancel the first two terms, leaving

$$\sin 135^\circ = \frac{\sqrt{2}}{2}.$$

Hence the exact value is

$$\boxed{\frac{\sqrt{2}}{2}}.$$

4. Let

$$f(x) = x - 2 \sin x \quad (0 \leq x \leq 2\pi).$$

Without calculus, determine how many solutions the equation $f(x) = \pi$ has, and locate each solution exactly if possible.

We solve

$$x - 2 \sin x = \pi,$$

or

$$x - \pi = 2 \sin x.$$

Let

$$t = x - \pi.$$

Then $x = \pi + t$, where $t \in [-\pi, \pi]$, and since

$$\sin(\pi + t) = -\sin t,$$

the equation becomes

$$t + 2 \sin t = 0.$$

Clearly $t = 0$ is a solution, giving

$$x = \pi.$$

We now show it is the only solution. Observe:

- If $0 < t \leq \pi$, then $\sin t \geq 0$, so

$$t + 2 \sin t > 0.$$

Hence no positive $t \neq 0$ works.

- If $-\pi \leq t < 0$, write $t = -u$ with $u > 0$. Then

$$t + 2 \sin t = -u - 2 \sin u < 0,$$

since $u > 0$ and $\sin u \geq 0$ for $0 \leq u \leq \pi$. Hence no negative t works.

So the only solution is

$$\boxed{x = \pi}.$$

Therefore the equation $f(x) = \pi$ has exactly

$$\boxed{1}$$

solution.

5. Find the least positive period of

$$g(x) = \sin\left(\frac{\pi x}{2}\right) + \cos\left(\frac{\pi x}{3}\right).$$

The period of

$$\sin\left(\frac{\pi x}{2}\right)$$

is T_1 such that

$$\frac{\pi}{2}T_1 = 2\pi,$$

so

$$T_1 = 4.$$

The period of

$$\cos\left(\frac{\pi x}{3}\right)$$

is T_2 such that

$$\frac{\pi}{3}T_2 = 2\pi,$$

so

$$T_2 = 6.$$

A common period of the sum must be a common multiple of 4 and 6. The least positive one is

$$\text{lcm}(4, 6) = 12.$$

Hence the least positive period is

$$\boxed{12}.$$

6. For real x , define

$$h(x) = \cos x + \cos\left(x + \frac{2\pi}{3}\right) + \cos\left(x + \frac{4\pi}{3}\right).$$

Evaluate $h(x)$, and then use the same idea to evaluate

$$\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right).$$

Use angle addition:

$$\cos\left(x + \frac{2\pi}{3}\right) = \cos x \cos \frac{2\pi}{3} - \sin x \sin \frac{2\pi}{3} = -\frac{1}{2}\cos x - \frac{\sqrt{3}}{2}\sin x,$$

$$\cos \left(x + \frac{4\pi}{3} \right) = \cos x \cos \frac{4\pi}{3} - \sin x \sin \frac{4\pi}{3} = -\frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x.$$

Adding with $\cos x$,

$$h(x) = \cos x - \frac{1}{2} \cos x - \frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x + \frac{\sqrt{3}}{2} \sin x = 0.$$

Therefore

$$\boxed{h(x) = 0 \text{ for all } x}.$$

Now similarly,

$$\sin \left(x + \frac{2\pi}{3} \right) = \sin x \cos \frac{2\pi}{3} + \cos x \sin \frac{2\pi}{3} = -\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x,$$

$$\sin \left(x + \frac{4\pi}{3} \right) = \sin x \cos \frac{4\pi}{3} + \cos x \sin \frac{4\pi}{3} = -\frac{1}{2} \sin x - \frac{\sqrt{3}}{2} \cos x.$$

Adding with $\sin x$,

$$\sin x - \frac{1}{2} \sin x - \frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x - \frac{\sqrt{3}}{2} \cos x = 0.$$

So

$$\boxed{\sin x + \sin \left(x + \frac{2\pi}{3} \right) + \sin \left(x + \frac{4\pi}{3} \right) = 0}.$$

Thus both sums are identically zero.